To

The coordinator

SSIP Cell

KSKV Kachchh University

Bhuj – Kutch 370001

Date: 16/02/2026

Subject: Research Proposal Submission - SSIP Fund - KU/ SSIP/ Proposals /2025-2026/01

Dear Sir,

This email pertains to the SSIP Fund application; reference number KU/ SSIP/ Proposals /2025-2026/01. I, Vaghela Trishaba Sanjaysinh, a MSc CA&IT semester 6 student in the Department of Computer Science, hereby submit my research proposal for your consideration.

The attached document details my proposed research.

Thank you for your time and attention to this matter.

Yours Sincerely,

Vaghela Trishaba Sanjaysinh

MSc CA&IT Semester - 6

Department of Computer Science, KSKVKU Kachchh University, Bhuj – Kachchh

SSIP Research Proposal

1. Applicant Information

- **Name of Applicant**: Vaghela Trishaba Sanjaysinh
- **Designation:** MSC CA & IT (Sem - 6)
- **Institute Name:** Department Of Computer Science, KSKVKU Kachchh University, Bhuj – Kachchh
- **Date of Birth (DOB):** 03/08/2006
- **WhatsApp Number:** 82007 03233
- **Email Id:** vtrishaba@gmail.com
- **Corresponding Address:** Department of Computer Science, KSKV Kachchh University, Bhuj – Kutch

2. Proposal Details

- **Title of Proposal: KishanMitra -** AI Enabled Kishan Mitra Platform
- **Abstract**

This project proposes an AI-enabled IoT-based smart motor control system for agriculture and home irrigation management. The system integrates soil moisture sensors, water level monitoring, motor current sensing, and weather data with cloud-based analytics. It enables remote monitoring and automated irrigation scheduling to reduce water wastage, improve energy efficiency, and prevent motor damage. The proposed solution aims to support sustainable agriculture through intelligent automation.

- ## Introduction

Most irrigation systems still depend on manual motor operation, where water is supplied based on fixed timing or estimation rather than actual soil conditions, leading to water wastage, higher electricity consumption, and motor damage due to dry running or overload. Farmers also need to frequently visit fields to operate pumps, which is time-consuming and inconvenient. By using IoT sensors and AI analysis, real-time monitoring of soil moisture, water level, and motor parameters becomes possible, allowing irrigation only when required. The proposed system develops an AI-enabled smart motor control solution that automates irrigation,

protects equipment, optimizes resource usage, and supports sustainable agriculture while reducing manual effort.

- ## Literature Review

Traditional irrigation systems mainly depend on manual control and fixed scheduling, which often leads to water wastage and inefficient energy use [3]. Research in smart agriculture introduced soil moisture sensors and IoT-based monitoring systems to automate irrigation based on real-time field conditions [1]. These systems improved water efficiency but mostly relied on simple threshold-based control [3]. Recent studies focus on AI and machine learning models to predict irrigation needs using historical data and weather information, enhancing precision and sustainability [5]. Additionally, smart motor control systems with current sensing and dry-run protection have been developed to reduce equipment damage and energy loss [2]. However, limited research integrates AI-based irrigation prediction with intelligent motor protection in a single unified platform, which this project aims to address [4].

- ## Methodology

1. Data Collection

Sources of Data:

- Field-based data from soil moisture, water level, flow, motor current, and weather sensors.
- Historical agricultural and weather datasets for irrigation pattern analysis.
- User feedback from farmers regarding irrigation practices and motor usage issues.

Types of Data Collected:

- Soil moisture levels and environmental conditions (temperature, humidity).
- Water tank levels, flow rate, and motor current data.
- Basic farm details (crop type, land size, irrigation schedule).

Ethical Considerations:

- Ensure secure storage of collected data.

- Maintain user privacy and data confidentiality.

2. Data Preprocessing

- **Data Cleaning:** Handle missing sensor readings and remove abnormal values.
- **Normalization:** Scale sensor data for accurate AI model training.
- **Feature Selection:** Identify key parameters influencing irrigation needs and motor safety.

3. Model Development

- Develop AI-based irrigation prediction using regression and ensemble methods.
- Split dataset into training and testing sets for validation.
- Apply cross-validation and hyperparameter tuning to improve accuracy.

4. Evaluation & Deployment

- Evaluate performance using metrics such as MAE and RMSE.
- Conduct field testing to validate real-time performance.
- Deploy the final system with a cloud dashboard and mobile monitoring interface.

3. Financial Information

○ Expenditure (Detailed Budget for the Project)

(a) Equipment Costs

Equipment	**Estimated Cost (₹)**
Industrial IoT Motor Controller & Gateway	₹ 18,850
Edge Computing Device (Raspberry Pi)	₹ 6,500
Protection & Relay Systems	₹ 2,000
Subtotal (Equipment):	**₹ 27,350**

(b) Materials and Supplies

Item	Estimated Cost (₹)
Soil & Water Sensors	₹4,000
Flow & Electrical Sensors	₹2,000
Subtotal (Materials & Supplies):	**₹ 6,000**

(c) Software & Technology

Software / Technology	Estimated Cost (₹)
Cloud Hosting (1 Year)	₹40,000
Database & Server Maintenance	₹5,000
Cybersecurity & SSL	₹10,000
Subtotal:	**₹ 55,000**

(d) Data Collection & Miscellaneous

Item	Estimated Cost (₹)
Data Collection	₹2,000
Subtotal:	**₹ 2,000**

Total Budget

Category	Total Cost
Equipment	₹ 27,350
Materials	₹ 6,000
Software	₹ 55,000
Miscellaneous	₹ 2,000
Total:	**₹ 90,350** **Ninety thousand three hundred fifty only**

4. Project Planning

- **Work Plan**

Phase	**Duration**	**Key Activities & Milestones**
Phase 1	Month 1 – 2	Literature review, finalize system requirements, select sensors and hardware components, design system architecture.
Phase 2	Month 2 – 4	Hardware setup: connect soil moisture, water level, flow and current sensors with ESP32 and gateway. Begin basic data collection.
Phase 3	Month 4 – 6	Develop IoT communication (MQTT & cloud database). Store and monitor real-time sensor data through dashboard.
Phase 4	Month 6 – 8	Develop AI model for irrigation prediction and motor automation based on collected data and weather conditions.
Phase 5	Month 8 – 10	Field testing on farm/home tanks, optimize motor protection features (dry run,

		overload, auto-cutoff). Improve accuracy.
Phase 6	Month 10 – 12	Final system deployment, mobile monitoring interface, performance evaluation,

- **Objectives**

AI-Enabled Smart Irrigation System: To develop a functional IoT-based prototype that monitors soil moisture, water level, and motor parameters to automate irrigation efficiently.

Improved Prediction Accuracy: To implement an AI model that optimizes irrigation scheduling and reduces water and energy wastage compared to traditional methods.

Motor Protection & Safety: To integrate dry-run and overload protection features to prevent motor damage.

Remote Monitoring: To enable real-time monitoring and control through a cloud-based platform.

Sustainability & User Acceptance: To promote water conservation and evaluate user satisfaction through field testing.

5. References

1] Liakos, K. et al. (2023). Machine Learning in Smart Irrigation: Recent Advances. Agricultural Water Management.

2] Sharma, A. et al. (2022). IoT-Based Smart Agriculture Monitoring Systems: A Review. IEEE Internet of Things Journal.

3] Kumar, P. & Singh, R. (2024). AI-Driven Precision Irrigation Using Edge Computing. Computers and Electronics in Agriculture.

4] FAO (2023). Digital Agriculture for Water Efficiency. Food and Agriculture Organization Report.

5] Patel, H. et al. (2022). Smart Motor Control and Energy Optimization in Agriculture. IEEE Access.

6. Details of Mentor

- **Name of Mentor:** Dr. MAHESHKUMAR D. MULANI
- **Designation:** Assistant Professor, Head of the Department
- **Name of the Institute:** Department of Computer Science, KSKV Kachchh University, Bhuj - Kutch
- **WhatsApp Number of Mentor:** +91 94094 93497
- **Email ID of Mentor:** mdm@kskvku.ac.in

7. Declaration

The information provided is true and the work will be carried out ethically under institutional guidelines.

Signature of Student: _____

Name and Signature of Faculty Mentor: Dr. Mahesh D. Mulani _____

Head of Department: Dr. Mahesh D. Mulani

Vaghela Trishaba Sanjaysinh

Present Designation: MSC CA & IT (Sem-6)

Address:

Department Of Computer Science
KSKVKU Kachchh University,
Bhuj – Mundra Road, Near Changleshwar Mahadev Temple
City – Bhuj, District – Kachchh, State – Gujarat,
Country - India, Pin Code – 370001

Contact Number: +91 82007 03233

Email: vtrishaba@gmail.com

PERSONAL DETAILS:

1. **Date Of Birth:** August 3,2006
2. **Gender:** Female
3. **Blood Group:** B Positive
4. **Language Knowledge:** Gujrati, Hindi, English
5. **Instagram:** https://www.instagram.com/_dell_diamond_3
6. **LinkedIn:** https://www.linkedin.com/in/vaghela-trishaba-05b457289

EDUCATIONAL QUALIFICATIONS:

Sr No.	Degree	Passing Year	University/Institute
1.	B.Sc. (CA&IT)	2026	K.S.K.V. Kachchh University, Bhuj
2.	HSC	2023	S.N.K.K.V, Bhavnagar
3.	SSC	2021	S.N.K.K.V, Bhavnagar

WORK EXPERIENCE:

WRTeam (Backend Developer; December 2025 - Present)

Worked as a Backend Developer Internship at WRTeam, where I developed and maintained web application using Laravel and Next.js across multiple projects.

PROJECTS:

1. Developed a School Management Platform with strip payment gateway integration for secure online fee transactions using Laravel.
2. Built a Mall Management System in Next.js, handling core features and overall system functionality.

OTHER ACHIVEMENTS:

1. Won a medal for creating website during academic competition.
2. Winner of hackathon organized by MBA department.

भारत सरकार

आधार

ભારત સરકાર
Government of India

ભારતીય વિશિષ્ટ ઓળખાણ પ્રાધિકરણ
Unique Identification Authority of India

નામાંકન ક્રમ સંખ્યા/ Enrolment No.: 2709/14030/00211

To
ત્રીશાબા સંજયસિંહ વાઘેલા
Trishaba Sanjaysinh Vaghela
VTC: Khoda,
PO: Khoda,
Sub District: Sanand,
District: Ahmedabad,
State: Gujarat,
PIN Code: 382170,
Mobile: 9974373530

Signature Not Verified
Digitally signed by DS Unique Identification Authority of India
OS
Date: 2026.01.29 12:37:42 IST

તમારો આધાર નંબર / Your Aadhaar No. :
8415 3768 1761
VID : 9108 3458 5719 9197

મારો આધાર, મારી ઓળખ

ભારત સરકાર
Government of India

આધાર

Aadhaar no. issued: 29/06/2016

ત્રીશાબા સંજયસિંહ વાઘેલા
Trishaba Sanjaysinh Vaghela
જન્મ તારીખ/DOB: 03/08/2006
સ્ત્રી/ FEMALE

આધાર એ ઓળખનો પુરાવો છે, નાગરિકતા અથવા જન્મ તારીખનો નહીં. તેનો ઉપયોગ માત્ર ચકાસણી (ઓનલાઇન પ્રમાણીકરણ અથવા ક્યુઆર કોડ/ઓફલાઇન એકસએમએલનું સ્કેનીંગ) સાથે જ થવો જોઇએ.
Aadhaar is proof of identity, not of citizenship or date of birth. It should be used with verification (online authentication, or scanning of QR code / offline XML).

8415 3768 1761